

# MIN-JONG BONG

Ph.D. Candidate in Inorganic Chemistry · Korea University · Advisor: Prof. Ho-Jin Son

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## RESEARCH INTEREST

Ph.D. candidate in Inorganic Chemistry at Korea University (expected February 2027), specializing in the design of single-molecule Mn(I) catalysts for selective CO<sub>2</sub> reduction. Research centers on secondary coordination sphere engineering of well-defined molecular complexes — inspired by formate dehydrogenase enzymes — integrating experimental (operando FTIR, CV, XRD) and computational (DFT; ORCA) approaches with Python-based automation of computational workflows. Key results include a TON of ~300 with >94% formate selectivity, published in Chemical Science (2026, first author).

**Keywords:** Organometallic Synthesis, CO<sub>2</sub> Reduction, Photocatalysis, Secondary Coordination Sphere, Operando FTIR, DFT

## EDUCATION

**Ph.D. candidate, Inorganic Chemistry**

Mar 2024 – Present

Korea University · Department of Advanced Materials Chemistry · Advisor: Prof. Ho-Jin Son

**M.S., Inorganic Chemistry**

Mar 2022 – Feb 2024

Korea University · Department of Advanced Materials Chemistry · Advisor: Prof. Ho-Jin Son

**B.S., Advanced Materials Chemistry**

Mar 2018 – Feb 2022

Korea University (Sejong)

## PUBLICATIONS

- [1] Secondary Sphere Impact on Organometallic Catalysts in Photochemical CO<sub>2</sub> Reduction: A Comparative Study of Tethering Ligand Lengths with Secondary Coordination**  
Min-Jong Bong, Wonjung Lee, Junhyeok Seo\*, Ho-Jin Son\*  
In preparation first author
- [2] Unveiling Key Factors Governing the Catalytic Performance of Half-Metallocene Complexes for Photochemical CO<sub>2</sub> Reduction and Hydrogen Production: Relativistic Effects and Charge Delocalization**  
Daehan Lee†, **Min-Jong Bong**†, Seunghwan Cha, Sangheon Jeong, Jung-Min Ji, Won-Jo Jung, Nady A. Fathy, Ho-Jin Son\*  
*Inorganic Chemistry Frontiers* · Under revision · *Royal Society of Chemistry* co-first author †
- [3] A secondary-sphere proton channel accelerating metal-hydride formation in Mn(I) catalysts for selective CO<sub>2</sub>-to-formate conversion**  
**Min-Jong Bong**†, Wonjung Lee†, Daehan Lee, Hyunuk Kim, Junhyeok Seo\*, Ho-Jin Son\*  
*Chemical Science* · 2026 · *Royal Society of Chemistry* · DOI: 10.1039/D5SC09412G first author
- [4] Efficient and Durable Photochemical CO<sub>2</sub> Reduction by TiO<sub>2</sub>-Immobilized Metal Porphyrin Catalysts**  
Hyeongu Kang, Daehan Lee, Sangheon Jeong, Seunghwan Cha, **Min-Jong Bong**, Myung Jae Lee, Won-Jo Jung, Ho-Jin Son\*  
*ChemSusChem*, 19, e202501889 · 2026 · *Wiley-VCH / Chemistry Europe* · DOI: 10.1002/cssc.202501889

[5] **Organic phosphorescence in Pt(II)-complexes linked to organic chromophores for blue-emitting organic light-emitting diodes**

Seong Woon Jeong<sup>†</sup>, Hyung Joo Lee<sup>†</sup>, Bum June Park, **Min-Jong Bong**, Daehan Lee, Taekyung Kim\*, Chul Hoon Kim\*, Ho-Jin Son\*

*Communications Chemistry*, 8, 140 · 2025 · *Nature Portfolio* · DOI: 10.1038/s42004-025-01533-y

## RESEARCH PROJECTS

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### Tether Length Modulation of Secondary Coordination Layer

2024 – Present

Systematic variation of –OH/–OMe pendant length at 6,6'-bipy positions of Mn(I) complexes. Best performer Mn-DiMeOMe: TON 368, formate selectivity 97.8%.

[Photocatalysis](#) · [CV](#) · [Operando FTIR](#) · [DFT](#) · [XRD](#)

### Proton Relay Engineering in Mn(I) Complexes

2022 – 2025

Ethylene-bridged pendants mimic formate dehydrogenase. Mn-bpydiOMe: TON ~300, formate selectivity >94%. Published in *Chem. Sci.* 2026.

[CO<sub>2</sub> Reduction](#) · [Operando FTIR](#) · [DFT](#) · [Mechanistic Study](#)

## SKILLS & METHODS

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|                       |                                                                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Technical:</b>     | Organic & organometallic synthesis, Glovebox & Schlenk techniques, Air-sensitive compound handling, Photocatalysis, 3D Graphic Design (Adobe Dimension)                        |
| <b>Instrumental:</b>  | Operando FTIR, Cyclic Voltammetry, UV-Vis Spectroscopy, Quantum Yield Measurement, Single-crystal XRD, Structure refinement (SHELXTL/SHELXL), NMR, ESI-MS, GC-MS, GC-TCD, HPLC |
| <b>Computational:</b> | DFT (ORCA, Gaussian), CPCM solvation, Reaction pathway mapping, Python (computational data automation & visualization), AI-assisted computational workflow (Claude Code)       |

## TEACHING & MENTORING

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### General Chemistry Laboratory 1

**Lecturer**

Spring 2026

Serving as course instructor for general chemistry laboratory.

### Inorganic Chemistry Laboratory

**Lecturer**

Fall 2026

Serving as course instructor for inorganic chemistry laboratory.

### CURT Undergraduate Research

**Mentoring**

Feb – Jul 2024

Mentored undergraduates in research design and execution.

### General Chemistry Laboratory 1

**TA**

Spring 2022, 2023, 2024, 2025

Supervised lab sessions for general chemistry.

### General Chemistry Laboratory 2

**TA**

Fall 2022, 2024

Supervised lab sessions for general chemistry.

### Inorganic Chemistry Laboratory

**TA**

Fall 2023, 2024, 2025

Instructed inorganic synthesis experiments.

## HONORS & AWARDS

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### Excellence in Poster Presentation Award

**2025**

Inorganic Chemistry Division, Korean Chemical Society

*Second Coordination Sphere Engineering for Efficient and Selective CO<sub>2</sub>-to-Formate Conversion of Mn(I) Bipyridyl Complexes: Steric and Brønsted Acidity/Basicity Influence of Alcohol and Alkyloxy Pendants and Its Mechanistic Investigation*

**Excellent Research Award**

2021

CURT Research Program, Korea University

*A Hybrid Ru/TiO<sub>2</sub> Catalyst for Photocatalytic CO<sub>2</sub> to CO/Formate Conversion*